

Supplementary Information

Evaluation Choices Shape Apparent Quantum Kernel Robustness under Distribution Shift

Roberto Fernández-Barrios, Iker Pastor-López,
Asier González-Santocildes, and Pablo García Bringas

Faculty of Engineering, University of Deusto, Bilbao, Spain

Supplementary Results: Scope

This document contains extended results, complete sensitivity tables, and diagnostics supporting the main article. No procedure is defined exclusively here: all data construction, preprocessing, candidate grids, selection rules, inference, external acquisition, geometry calculations, and finite-shot procedures are reported in the main manuscript Methods. Machine-readable values and frozen input hashes are included in the immutable release artifact.

Supplementary Results: Cross-generation sensitivity map

The ten rows remain in their prespecified S1–S10 order rather than being ranked by outcome. Every row includes all eight fixed security scenario-groups. Scenario-groups are averaged within EMBER, UNSW-NB15, and ToN-IoT before the three source datasets receive equal weight. The corrected S10 values in Table 1 are identical to the confirmatory endpoint in the main article. This display compares complete analysis specifications across two result generations; it is not a factorial decomposition of their individual choices.

Supplementary Table 1: Prespecified cross-generation definitions and source-dataset-equal quantum-minus-classical OOD balanced-accuracy effects. “Oracle” selects and evaluates on the same OOD test labels; “ID-val” uses a disjoint in-distribution validation split. SVC regularization is fixed at $C = 1$ or selected by training-only cross-validation; this axis is not applicable to GPC. S1–S4 use the legacy generation and S5–S10 use v4, so their boundary does not isolate the effect of regularization. Values are descriptive fixed-case summaries, not per-axis causal effects.

ID	Selection	SVC C	Classical reference	Budget	SVC	GPC
S1	OOD oracle	Fixed	Linear/RBF	Native	+0.0313	+0.0340
S2	Legacy ID	Fixed	Linear/RBF	Native	+0.0475	+0.0418
S3	OOD oracle	Fixed	Extended	Native	+0.0058	+0.0145
S4	Legacy ID	Fixed	Extended	Native	+0.0014	+0.0122
S5	OOD oracle	Train CV	Linear/RBF	Native	+0.0034	+0.0125
S6	ID-val	Train CV	Linear/RBF	Native	−0.0041	+0.0059
S7	OOD oracle	Train CV	Extended	Native	−0.0054	+0.0012
S8	ID-val	Train CV	Extended	Native	−0.0038	−0.0029
S9	OOD oracle	Train CV	Extended	Equal	−0.0032	+0.0038
S10	ID-val	Train CV	Extended	Equal	−0.0056	−0.0009

The source-dataset-equal range is 0.0531 for SVC and 0.0447 for GPC. This spread is not an interval: the specifications share data and candidates and deliberately target different estimands.

Supplementary Results: Candidate-budget sampling

Table 2 shows every group–classifier cell under all three prespecified sampling schemes. The primary blocked scheme mirrors the five-dimension block structure of the quantum pool; the other schemes match candidate count but not the same search-space structure. Their close values show that the controlled conclusion is not an artefact of the blocked draw.

Supplementary Table 2: Complete equal-budget sampling sensitivity. Entries are median quantum-minus-classical $P1'$ OOD differences over 5,000 classical-pool subsamples. “Blocked” is primary; “uniform” and “stratified” are sensitivities. Source-dataset-equal rows average scenario-group medians within the three source datasets. These resampling medians are distinct from the run-level expected-classical endpoint used for the conditional intervals in the main article.

Scenario	Clf.	Blocked	Uniform	Stratified
EMBER $m1$	SVC	−0.0056	−0.0054	−0.0054
	GPC	−0.0004	−0.0009	−0.0013
EMBER $m2$	SVC	−0.0071	−0.0071	−0.0071
	GPC	+0.0007	+0.0008	+0.0005
UNSW-DoS (campaign)	SVC	−0.0036	−0.0034	−0.0036
	GPC	−0.0072	−0.0072	−0.0071
UNSW-DoS (constructed)	SVC	−0.0046	−0.0030	−0.0028
	GPC	−0.0116	−0.0113	−0.0112
UNSW-Recon (campaign)	SVC	+0.0050	+0.0052	+0.0052
	GPC	+0.0046	+0.0047	+0.0049
UNSW-Recon (constructed)	SVC	−0.0090	−0.0089	−0.0089
	GPC	−0.0082	−0.0076	−0.0077
ToN-IoT (campaign)	SVC	−0.0147	−0.0149	−0.0143
	GPC	−0.0095	−0.0094	−0.0096
ToN-IoT (constructed)	SVC	−0.0011	−0.0008	−0.0011
	GPC	+0.0093	+0.0063	+0.0057
Source-dataset-equal	SVC	−0.00576	−0.00555	−0.00548
	GPC	−0.00185	−0.00232	−0.00253

The largest within-cell scheme range is 0.00353 for ToN-IoT constructed under GPC. The next largest is 0.00184 for UNSW-DoS constructed under SVC.

Supplementary Results: Effective-rank matching sensitivities

The predefined rank ratio is $\max(r_Q, r_C) / \min(r_Q, r_C)$. Table 3 reports all calipers for both observational matchers. With replacement, 75.2% of pairs survive the primary 1.25 caliper. One-to-one assignment has a smaller common support and becomes visibly poor without a caliper (95th-percentile ratio 11.45), making the overlap restriction scientifically material.

At the primary 1.25 caliper, both matchers yield a negative group median in all eight groups. The result is descriptive of the retained overlap, not a causal family estimate.

Supplementary Results: Geometry associations

Effective-rank association changes sign across regimes, whereas OOD kernel-target alignment is more consistently positive (Table 4). The source-dataset leave-one-out ordering correlations are 0.28 for EMBER, 0.22 for UNSW-NB15, and 0.40 for ToN-IoT when trained on the other two sources.

Supplementary Table 3: Pooled SVC effective-rank matching sensitivity. “Nearest” pairs each quantum candidate to its nearest classical rank with classical reuse; “one-to-one” uses minimum-cost assignment without reuse. Retained percentages use 64,800 possible quantum candidates as denominator. Δ is quantum minus classical OOD balanced accuracy.

Matcher	Caliper	Pairs (%)	Median ratio	95th pct. ratio	Median Δ	Q better
Nearest	1.10	27,928 (43.1)	1.040	1.093	−0.0054	29.4%
	1.25	48,708 (75.2)	1.082	1.226	−0.0056	29.7%
	1.50	60,647 (93.6)	1.112	1.390	−0.0061	28.9%
	2.00	64,408 (99.4)	1.123	1.542	−0.0070	27.8%
	none	64,800 (100)	1.124	1.578	−0.0071	27.6%
One-to-one	1.10	13,205 (20.4)	1.044	1.094	−0.0048	30.2%
	1.25	24,675 (38.1)	1.092	1.232	−0.0053	30.3%
	1.50	35,129 (54.2)	1.151	1.447	−0.0057	30.2%
	2.00	46,898 (72.4)	1.232	1.829	−0.0057	30.8%
	none	64,800 (100)	1.418	11.452	−0.0062	30.9%

Supplementary Table 4: Geometry–OOD association and primary nearest-rank pairing for SVC. “Partial” controls kernel family and scale. Matching uses replacement and rank ratio ≤ 1.25 ; Δ is the median quantum-minus-classical OOD difference. All quantities are observational.

Scenario	ρ rank	Partial	ρ KTA	Matched Δ	Q better
EMBER $m1$	+0.32	+0.36	+0.30	−0.0094	33%
EMBER $m2$	+0.80	+0.82	+0.63	−0.0089	22%
UNSW-DoS (campaign)	+0.44	+0.41	+0.64	−0.0089	27%
UNSW-DoS (constructed)	−0.14	−0.08	+0.39	−0.0057	29%
UNSW-Recon (campaign)	+0.06	+0.06	+0.22	−0.0047	37%
UNSW-Recon (constructed)	−0.15	−0.08	+0.37	−0.0061	30%
ToN-IoT (campaign)	+0.25	+0.27	+0.46	−0.0060	24%
ToN-IoT (constructed)	+0.65	+0.63	+0.59	−0.0013	35%

Supplementary Results: Descriptive oracle comparison

The initial EMBER evaluation selected its representative using OOD test labels and fixed SVC at $C = 1$. Expanding the reference family nearly removes the apparent gain even before target-label-free selection is imposed.

Supplementary Table 5: Descriptive EMBER oracle results under fixed $C = 1$ (18 correlated settings). Mean and median effects are quantum minus classical balanced accuracy. These values are not deployable estimates or independent replicates.

Selection	Classifier	Classical reference	Q wins	Mean Δ	Median Δ
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426
	SVC	extended	15/18	+0.0044	+0.0064
	GP	linear+RBF	15/18	+0.0283	+0.0339
	GP	extended	11/18	+0.0023	+0.0029
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794
	SVC	extended	18/18	+0.0149	+0.0152
	GP	linear+RBF	18/18	+0.0649	+0.0664
	GP	extended	18/18	+0.0192	+0.0166

Supplementary Results: Small-cluster interval sensitivity

Table 6 compares the primary pointwise conditional cluster- t interval with the prespecified BCa and leave-one-q-split diagnostics. The effective replication unit is five q-split clusters in every row. These

intervals are not simultaneous and are not used as a familywise-error-controlled classification of the eight cases. Differences between interval constructions do not change the qualitative fixed-case description, and an interval containing zero is not interpreted as equivalence. With only five clusters, the BCa interval is retained as a fragile sensitivity rather than treated as additional confirmatory evidence.

Supplementary Table 6: Pointwise conditional uncertainty sensitivity for all primary equal-budget group–classifier cells. Δ is quantum minus classical OOD balanced accuracy. The cluster- t and 9,999-draw BCa columns are intervals over the same five q-split cluster means ($n = 5$); LOO is the range after leaving out one q-split cluster. All are conditional on the fixed benchmark pools and design, not intervals over a dataset population. They are not simultaneous intervals, and the BCa column is fragile at this effective sample size.

Scenario	Clf.	Δ	95% cluster- t	95% BCa	LOO range
EMBER $m1$	GPC	-0.0009	[-0.0101, +0.0083]	[-0.0062, +0.0054]	[-0.0035, +0.0015]
	SVC	-0.0058	[-0.0340, +0.0223]	[-0.0226, +0.0110]	[-0.0120, +0.0014]
EMBER $m2$	GPC	+0.0027	[-0.0027, +0.0081]	[-0.0015, +0.0053]	[+0.0017, +0.0043]
	SVC	-0.0064	[-0.0137, +0.0009]	[-0.0107, -0.0018]	[-0.0081, -0.0049]
UNSW-DoS (campaign)	GPC	-0.0057	[-0.0120, +0.0005]	[-0.0084, +0.0004]	[-0.0078, -0.0046]
	SVC	-0.0033	[-0.0139, +0.0073]	[-0.0101, +0.0029]	[-0.0058, -0.0008]
UNSW-DoS (constructed)	GPC	-0.0108	[-0.0177, -0.0039]	[-0.0141, -0.0054]	[-0.0131, -0.0095]
	SVC	-0.0044	[-0.0119, +0.0031]	[-0.0087, +0.0005]	[-0.0067, -0.0026]
UNSW-Recon (campaign)	GPC	+0.0052	[-0.0019, +0.0123]	[-0.0001, +0.0089]	[+0.0038, +0.0072]
	SVC	+0.0050	[-0.0058, +0.0157]	[+0.0006, +0.0131]	[+0.0011, +0.0065]
UNSW-Recon (constructed)	GPC	-0.0075	[-0.0208, +0.0059]	[-0.0162, +0.0006]	[-0.0106, -0.0040]
	SVC	-0.0089	[-0.0146, -0.0031]	[-0.0123, -0.0053]	[-0.0102, -0.0076]
ToN-IoT (campaign)	GPC	-0.0090	[-0.0190, +0.0010]	[-0.0166, -0.0034]	[-0.0106, -0.0060]
	SVC	-0.0148	[-0.0268, -0.0029]	[-0.0218, -0.0068]	[-0.0180, -0.0129]
ToN-IoT (constructed)	GPC	+0.0109	[+0.0017, +0.0202]	[+0.0060, +0.0178]	[+0.0082, +0.0129]
	SVC	-0.0011	[-0.0049, +0.0028]	[-0.0033, +0.0013]	[-0.0019, -0.0003]

Supplementary Table 7: Five q -split cluster means underlying each primary pointwise conditional interval. Values are quantum-minus-expected-classical OOD balanced-accuracy differences under P1' and 60-candidate blocked matching. Lower-level model/master-seed and nested-size realizations are averaged within each displayed cluster; the five columns, not the individual pipeline realizations, are the effective replication units.

Scenario	Clf.	$q = 7$	$q = 42$	$q = 123$	$q = 999$	$q = 2024$
EMBER $m1$	SVC	-0.01393	+0.01886	-0.03483	-0.01476	+0.01558
	GPC	-0.00320	-0.00166	+0.00135	+0.00965	-0.01066
EMBER $m2$	SVC	-0.01221	-0.00951	-0.00064	+0.00052	-0.01010
	GPC	+0.00351	+0.00663	+0.00625	+0.00098	-0.00395
UNSW-DoS (campaign)	SVC	-0.00005	-0.01302	+0.00666	+0.00135	-0.01134
	GPC	-0.00595	-0.01046	-0.00637	-0.00859	+0.00270
UNSW-DoS (constructed)	SVC	-0.00466	-0.00689	-0.01185	+0.00469	-0.00340
	GPC	-0.01167	-0.01352	-0.01614	-0.00153	-0.01109
UNSW-Recon (campaign)	SVC	+0.00246	+0.00188	-0.00109	+0.02025	+0.00127
	GPC	+0.00998	+0.00553	-0.00281	+0.01107	+0.00233
UNSW-Recon (constructed)	SVC	-0.00424	-0.00367	-0.01388	-0.01128	-0.01122
	GPC	-0.01376	+0.00122	-0.00861	-0.02125	+0.00511
ToN-IoT (campaign)	SVC	-0.02155	-0.00193	-0.00700	-0.02105	-0.02260
	GPC	-0.02081	-0.00256	-0.00313	-0.01385	-0.00446
ToN-IoT (constructed)	SVC	+0.00217	-0.00268	-0.00324	+0.00247	-0.00403
	GPC	+0.02184	+0.01485	+0.00872	+0.00624	+0.00304

Supplementary Results: Quantum-map strata

Table 8 reports every fixed-case SVC effect after separating the 30 entangling-ZZ and 30 separable-product candidates. Each stratum is compared with the expected P1' winner over 5,000 blocked draws of six complete extended-classical blocks, also 30 candidates. The full pool uses 60 candidates per family. Intervals are pointwise and conditional on five q-split clusters; their inclusion of zero is not an equivalence decision.

Supplementary Table 8: Circuit-stratified quantum-minus-classical OOD balanced-accuracy effects under train-CV and P1'. Brackets for scenario-groups are pointwise 95% conditional cluster- t intervals ($n = 5$ q-split clusters), not simultaneous intervals. The source-dataset-equal row gives the mean followed by the range of its three constituent source-dataset means, not a confidence interval.

Scenario	All maps ($B = 60$)	Entangling ZZ ($B = 30$)	Product maps ($B = 30$)
EMBER $m1$	-0.0057 [-0.0340, +0.0225]	-0.0063 [-0.0279, +0.0153]	+0.0036 [-0.0132, +0.0204]
EMBER $m2$	-0.0065 [-0.0138, +0.0008]	-0.0053 [-0.0130, +0.0024]	-0.0135 [-0.0166, -0.0104]
UNSW-DoS (campaign)	-0.0033 [-0.0139, +0.0073]	-0.0129 [-0.0184, -0.0073]	-0.0016 [-0.0116, +0.0083]
UNSW-DoS (constructed)	-0.0045 [-0.0119, +0.0030]	-0.0149 [-0.0299, +0.0001]	-0.0007 [-0.0063, +0.0049]
UNSW-Recon (campaign)	+0.0049 [-0.0058, +0.0157]	-0.0014 [-0.0070, +0.0041]	+0.0057 [-0.0023, +0.0137]
UNSW-Recon (constructed)	-0.0088 [-0.0145, -0.0032]	-0.0171 [-0.0286, -0.0056]	-0.0025 [-0.0066, +0.0015]
ToN-IoT (campaign)	-0.0149 [-0.0270, -0.0028]	-0.0196 [-0.0303, -0.0089]	-0.0225 [-0.0347, -0.0102]
ToN-IoT (constructed)	-0.0011 [-0.0050, +0.0028]	-0.0013 [-0.0047, +0.0021]	-0.0009 [-0.0036, +0.0018]
Source-dataset-equal	-0.0057 [-0.0080, -0.0029]	-0.0093 [-0.0116, -0.0058]	-0.0055 [-0.0117, +0.0002]

For P1' selection, product maps account for 594/1,080 SVC winners and 348/1,080 GPC winners. Under same-test OOD selection, the corresponding counts are 650/1,080 and 528/1,080. Complete counts by individual map are shown in Table 9 and provided in `quantum_winner_composition.csv`.

Supplementary Table 9: Quantum-map composition of the within-family winner. Each model-selector column contains 1,080 fixed security runs. Counts and percentages describe which configuration maximizes the specified selection metric within the quantum family; they do not compare its OOD accuracy with the classical family. P1' uses disjoint ID validation, whereas the OOD oracle reuses the shifted-test labels.

Map	Stratum	P1' SVC	P1' GPC	OOD-oracle SVC	OOD-oracle GPC
ZZ, 1 rep.	entangling ZZ	290 (26.9%)	353 (32.7%)	282 (26.1%)	279 (25.8%)
ZZ, 2 reps.	entangling ZZ	196 (18.1%)	379 (35.1%)	148 (13.7%)	273 (25.3%)
Pauli-X/Z, 1 rep.	separable product	315 (29.2%)	171 (15.8%)	353 (32.7%)	246 (22.8%)
Z, 2 reps.	separable product	279 (25.8%)	177 (16.4%)	297 (27.5%)	282 (26.1%)

Supplementary Results: Within-v4 factorial

The q1000 factorial holds the v4 data generation, split roles, kernel inventory, and preprocessing fixed while crossing regularization, selector, classical reference, and budget mode. Table 10 reports all 16 three-source-dataset-equal endpoints. The fixed- C , same-test-oracle, customary-reference, native-budget corner is +0.0131; the train-CV, ID-validation, extended-reference, equal-count corner is -0.0047. These endpoints are fixed-case summaries, not population effects.

Table 11 summarizes paired axis changes and pairwise differences-in-differences. Each axis row averages its paired change over the other eight factorial cells; each interaction row averages over the four cells of the remaining two axes. The relatively large regularization-by-reference interaction shows why the axis averages are not interpreted as independent, additive, or causal contributions.

Supplementary Results: Network-field ablation

The frozen q1000 sensitivity removes port-, protocol-, and service-related fields before fitting the shared train-only embedding. The ablation changes absolute task difficulty, especially for the

Supplementary Table 10: Within-v4 q1000 SVC factorial. Entries are three-source-dataset-equal quantum-minus-classical OOD balanced-accuracy effects. “Native” uses the complete family pools; “equal count” block-samples the larger family. Positive values favour the fidelity-kernel family.

Regularization and selector	Customary native	Customary equal	Extended native	Extended equal
$C = 1$, OOD oracle	+0.0131	+0.0094	−0.0010	+0.0021
$C = 1$, ID validation	+0.0118	+0.0118	−0.0079	−0.0068
Train-CV, OOD oracle	+0.0018	−0.0012	−0.0066	−0.0040
Train-CV, ID validation	−0.0074	−0.0084	−0.0040	−0.0047

Supplementary Table 11: Descriptive factorial axis changes and pairwise interactions. Changes are level-two minus level-one quantum-minus-classical effects, so a negative value is a shift towards the classical family. Interactions are differences-in-differences. No population inference, causal attribution, or familywise decision rule is attached to these finite-design summaries.

Finite-design summary	Mean change
Regularization: $C = 1 \rightarrow$ train-CV	−0.0084
Selection: OOD oracle $\rightarrow$ ID validation	−0.0037
Reference: customary $\rightarrow$ extended	−0.0079
Budget: native $\rightarrow$ equal count	−0.0002
Interaction: reference $\times$ budget mode	+0.0034
Interaction: regularization $\times$ budget mode	−0.0007
Interaction: regularization $\times$ reference	+0.0139
Interaction: regularization $\times$ selection	+0.0000
Interaction: selection $\times$ budget mode	+0.0001
Interaction: selection $\times$ reference	+0.0004

ToN-IoT campaign shift, and can affect the two kernel families differently. Table 12 therefore reports original, ablated, and paired-change effects separately.

Supplementary Table 12: Port/protocol/service-field ablation under train-CV, P1’, and blocked 60-candidate matching (SVC). Scenario-group brackets are pointwise 95% conditional cluster- t intervals over five q-split clusters and are not simultaneous. The final-row bracket is the range of the UNSW-NB15 and ToN-IoT source-dataset changes, not a confidence interval.

Scenario	Original effect	Ablated effect	Ablated minus original
UNSW-DoS (campaign)	−0.0021 [−0.0208, +0.0167]	−0.0108 [−0.0231, +0.0016]	−0.0087 [−0.0234, +0.0061]
UNSW-DoS (constructed)	+0.0014 [−0.0132, +0.0161]	+0.0008 [−0.0264, +0.0280]	−0.0006 [−0.0234, +0.0221]
UNSW-Recon (campaign)	−0.0028 [−0.0275, +0.0220]	+0.0072 [−0.0091, +0.0234]	+0.0099 [−0.0268, +0.0466]
UNSW-Recon (constructed)	−0.0138 [−0.0284, +0.0007]	−0.0074 [−0.0289, +0.0141]	+0.0065 [−0.0186, +0.0315]
ToN-IoT (campaign)	−0.0007 [−0.0111, +0.0097]	−0.0801 [−0.1717, +0.0114]	−0.0794 [−0.1753, +0.0165]
ToN-IoT (constructed)	−0.0029 [−0.0095, +0.0037]	−0.0129 [−0.0516, +0.0259]	−0.0100 [−0.0459, +0.0259]
Two-source-dataset-equal	−0.0031	−0.0245	−0.0215 [−0.0447, +0.0018]

Supplementary Results: Circuit resources

Supplementary Table 13: Complete logical circuit audit. Counts follow Qiskit 2.3.1 decomposition to **rz**, **sx**, **x**, and **cx** at optimization level 0 without device routing. “Map” is one feature-map state preparation; “fidelity” is its compute–uncompute template for one pair of inputs. One-qubit counts exclude barriers and measurements. These compiler-conditional logical counts are not device costs.

Map	Qubits	Map depth	Map 1q	Map CX	Fidelity depth	Fidelity 1q	Fidelity CX
ZZ, 1 rep.	4	19	22	12	38	44	24
	6	31	39	30	62	78	60
	8	43	60	56	86	120	112
	10	55	85	90	110	170	180
	12	67	114	132	134	228	264
ZZ, 2 reps.	4	35	44	24	70	88	48
	6	53	78	60	106	156	120
	8	71	120	112	142	240	224
	10	89	170	180	178	340	360
	12	107	228	264	214	456	528
Pauli-X/Z, 1 rep.	4	11	44	0	22	88	0
	6	11	66	0	22	132	0
	8	11	88	0	22	176	0
	10	11	110	0	22	220	0
	12	11	132	0	22	264	0
Z, 2 reps.	4	8	32	0	16	64	0
	6	8	48	0	16	96	0
	8	8	64	0	16	128	0
	10	8	80	0	16	160	0
	12	8	96	0	16	192	0

Supplementary Results: External fixed tasks

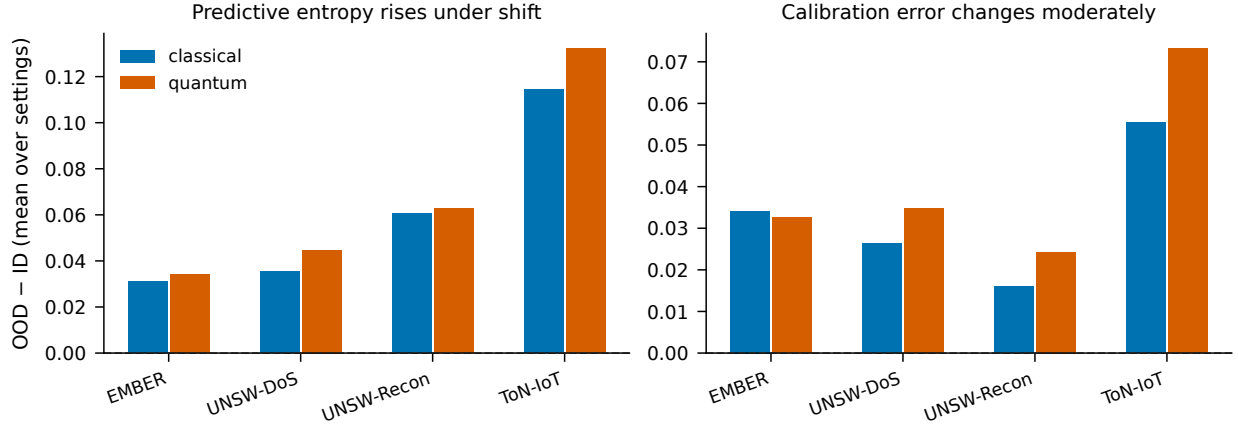
All 30 prospectively specified task–size–seed units passed the acquisition, split-integrity, nesting, and leakage audits. They contain 37,800 split-level results. Under SVC, size-equal controlled effects for College Scorecard, diabetes readmission, and ACS income are -0.0160 , -0.0066 , and -0.0131 ; their protocol contractions are $+0.0393$, $+0.0105$, and $+0.0126$. The task-equal controlled effect is -0.0119 (95% conditional seed-cluster interval $[-0.0205, -0.0033]$, $n = 5$ seeds per task), and contraction is $+0.0208$ ($[+0.0113, +0.0307]$). Under GPC, the task-equal controlled effect is $+0.0077$ ($[-0.0015, +0.0183]$) and contraction is $+0.0324$ ($[+0.0223, +0.0427]$). These fixed-task results corroborate protocol sensitivity, not a universal family ordering.

Supplementary Results: Predictive uncertainty

Predictive entropy rises from ID to OOD for both honestly selected families in every scenario-group. Expected calibration error changes moderately and comparably (Figure 1). Together with the controlled accuracy results, these diagnostics provide no consistent evidence in the fixed cases of a calibration benefit unique to the selected fidelity kernels. Entropy increase alone is not evidence of good calibration.

Supplementary Results: Repeated finite-shot sensitivity

Figure 2 displays all 2,880 finite-shot evaluations: eight fixed exact Gram matrices, four shot counts, 30 measurement replicates, and three PSD-handling conditions. The bands make the conditional Monte Carlo spread visible rather than treating diversity across data configurations as measurement replication. The independent-square heuristic and coherent train-based Nyström extension often



Supplementary Figure 1: Mean ID-to-OOD change in predictive entropy (left) and 15-bin expected calibration error (right) for the P1'-selected configuration of each family. Bars aggregate fixed scenario-groups by source dataset and are descriptive; no population interval is implied.

narrow or shift the distribution but do not impose a uniform direction. Exact-statevector OOD accuracy is approached unevenly because each sampled training matrix reselects a discrete SVC C .

Across the eight group medians, the perturbed/exact effective-rank ratio declines from 1.93 at 128 shots to 1.13 at 8,192 shots. OOD-KTA under the Nyström extension is estimator-dependent because its OOD square is derived from OOD-train features rather than independently sampled; it is not treated as a mechanism or compared directly with the other KTA estimators. Retained rank, eigenvalue tolerance, blockwise Frobenius changes, negative-spectrum fractions, selected C , and every replicate endpoint are provided in `results/v6/shots_mc/`.

Supplementary Table 14: Projected sampling resources for the repeated finite-shot sensitivity. One case-replicate contains 1,624,250 distinct fidelity parameterizations. “All eight” multiplies one shot level by eight fixed cases and 30 measurement replicates; “entangling four” retains only the four selected ZZ cases. The other four selected configurations are separable-product maps whose direct classical factorization can bypass circuit sampling. The three PSD conditions reuse these measurements. Counts exclude device routing, calibration, mitigation, queueing, and repeated-job overhead; no hardware execution was performed.

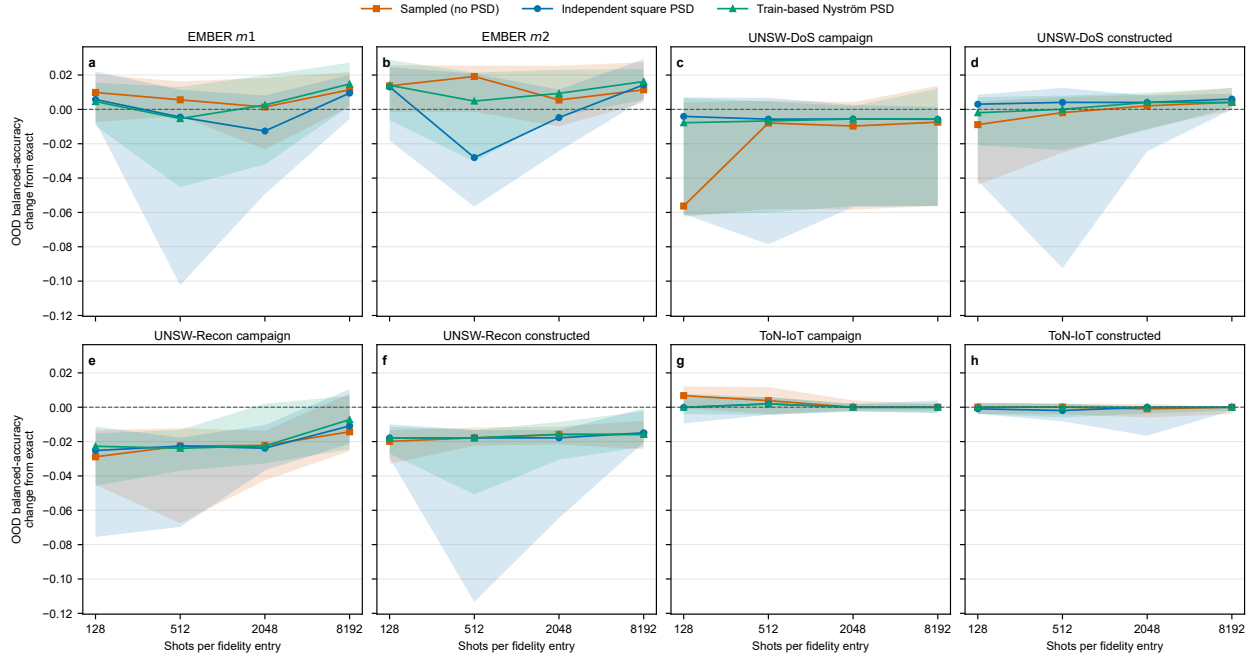
Shots/fidelity	Shots/case-replicate	Estimation instances/all eight	Shots/all eight	Shots/entangling four
128	207,904,000	389,820,000	49,896,960,000	24,948,480,000
512	831,616,000	389,820,000	199,587,840,000	99,793,920,000
2,048	3,326,464,000	389,820,000	798,351,360,000	399,175,680,000
8,192	13,305,856,000	389,820,000	3,193,405,440,000	1,596,702,720,000
All four	17,671,840,000	1,559,280,000	4,241,241,600,000	2,120,620,800,000

Supplementary Results: Machine-readable result map

The canonical v0.7.0 release gates are

```
python scripts/analysis/validate_v6_artifacts.py
pytest tests -q
```

The corrected three-source inference, budget schemes, rank sensitivities, and finite-shot outputs are under `results/v6/`; frozen legacy/v4/v5 artifacts remain unchanged. The development repository is https://github.com/roberto-fernandez-barrios/kernel_shift_framework, and the concept archive is <https://doi.org/10.5281/zenodo.19147649>. The exact v0.7.0 submission artifact is identified by the immutable version DOI <https://doi.org/10.5281/zenodo.21676563>.



Supplementary Figure 2: Repeated finite-shot OOD change from the exact fixed-configuration endpoint. Panels show one fixed scenario-group Gram matrix. Lines are medians over 30 independently seeded measurement replicates and shaded regions are central 95% Monte Carlo intervals ($n = 30$) conditional on that matrix. Orange squares use the sampled indefinite matrix; blue circles use independent PSD projections of train and OOD squares while leaving rectangular blocks unchanged; green triangles use the coherent train-eigenspace Nyström extension. The model includes fidelity sampling only, not device noise.